GiSMO - Factorisations

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Expansion

$(3x+4)(5x+1)$

$= 15x^2 + \underline{3x + 20x} + 4$

$= 15x^2 + 23x + 4$

$(a+b)(c+d) = ac + ad + bc + bd$

Factorisation = Reverse of expansion

$15x^2 + 23x + 4 = (3x+4)(5x+1)$

guess (reasonably) and check!!

1) $a^2 + 2a + 1 = (a+1)(a+1) = (a+1)^2$

$\sqrt{100^2 + 200 + 1} = \sqrt{(100+1)^{\cancel{2}}} = 101$

2) $a^2 - 4a + 4 = (a-2)(a-2) = (a-2)^2$

$$\begin{array}{cc|c} a & -2 & -2a \\ a & -2 & -2a \end{array}$$

3) $a^2 - 5a + 6 = (a-2)(a-3)$

$$\begin{array}{cc|c} a & -2 & -2a \\ a & -3 & -3a \\ \hline a^2 & 6 & -5a \end{array}$$

4) $2a^2 + 5a - 3 = (2a-1)(a+3)$

$$\begin{array}{cc|c} 2a & -1 & -a \\ a & 3 & 6a \\ \hline 2a^2 & -3 & 5a \end{array}$$

5) $a^2 - b^2 = (a-b)(a+b) = a^2 + \cancel{ab} - \cancel{ab} - b^2$

(difference of squares) $= a^2 - b^2$

6) $a^3 - b^3 \rightarrow$ guess and check alone is difficult!! :(

(difference of cubes)

1. guess and check, but <u>continue</u> when wrong

$a^3 - b^3 \neq (a-b)(a^2+b^2) \rightarrow$ example of wrong guess

$= a^3 + ab^2 - a^2b - b^3$

$$(a-b)(a^2+b^2) = a^3 + ab^2 - a^2b - b^3$$

$$\begin{aligned} a^3 - b^3 &= (a-b)(a^2+b^2) - ab^2 + a^2b \\ &= (a-b)(a^2+b^2) + ab(a-b) \\ &= (a-b)(a^2+b^2+ab) \checkmark \end{aligned}$$

2. "smarter" method: start with an identity you already know

(actually similar to above)

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$\begin{aligned} a^3 - b^3 &= (a-b)^3 + 3a^2b - 3ab^2 \\ &= (a-b)^3 + 3ab(a-b) \\ &= (a-b)((a-b)^2 + 3ab) \\ &= (a-b)(a^2+ab+b^2) \checkmark \end{aligned}$$

7) $a^3 + b^3$ (sum of cubes)

1. "smarter" method

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$\begin{aligned} a^3 + b^3 &= (a+b)^3 - 3a^2b - 3ab^2 \\ &= (a+b)^3 - 3ab(a+b) \\ &= (a+b)((a+b)^2 - 3ab) \\ &= (a+b)(a^2 - ab + b^2) \checkmark \end{aligned}$$

2. even smarter method :)

$a^3 - c^3 = (a-c)(a^2+ac+c^2)$

$\hookrightarrow$ plug in $c=-6$

$a^3 - (-6)^3 = (a-(-6))(a^2+a(-6)+(-6)^2)$

$a^3 + 6^3 = (a+6)(a^2-a6+6^2)$

8) $a^4 - 6^4 = (a^2)^2 - (6^2)^2$

$= (a^2-6^2)(a^2+6^2)$

$= (a-6)(a+6)(a^2+6^2)$

9) $a^4 + a^2 + 1$

$(a^2+1)(a^2+1) = a^4 + a^2 + a^2 + 1$

$a^4 + a^2 + 1 = (a^2+1)^2 - a^2$

$= (a^2+1-a)(a^2+1+a)$ ✓

10) $a^4 + 4b^4$ (Sophie Germain)

$(a^2+2b^2)(a^2+2b^2) = a^4 + 4a^2b^2 + 4b^4$

$a^4 + 4b^4 = (a^2+2b^2)^2 - 4a^2b^2$

$= (a^2+2b^2)^2 - (2ab)^2$

$= (a^2+2b^2-2ab)(a^2+2b^2+2ab)$ ✓

11) $a^4 - 3a^2b^2 + b^4$

$(a^2+b^2)(a^2+b^2) = a^4 + 2a^2b^2 + b^4$

$a^4 + b^4 = (a^2+b^2)^2 - 2a^2b^2$

$a^4 - 3a^2b^2 + b^4 = (a^2+b^2)^2 - 5a^2b^2$ ✗

$(a^2-b^2)(a^2-b^2) = a^4 - 2a^2b^2 + b^4$

$$a^4+b^4 = (a^2-b^2)^2+2a^2b^2$$

$$a^4-3a^2b^2+b^4 = (a^2-b^2)^2 - a^2b^2$$

$$= (a^2-b^2)^2 - (ab)^2$$

$$= (a^2-b^2-ab)(a^2-b^2+ab) \checkmark$$

12) $(a+b)(a-b) - 4(b+1)$

$$= a^2-b^2-4b-4$$

$$= a^2 - (b^2+4b+4)$$

$$= a^2 - (b+2)^2$$

$$= (a-b-2)(a+b+2) \checkmark$$

13) $a^2 - b^2 + 2(a+3b-4)$

$$= a^2-b^2+2a+6b-8$$

$$= a^2+2a-b^2+6b-8$$

$$= (a^2+2a+1)-1-b^2+6b-8$$

$$= (a+1)^2-b^2+6b-9$$

$$= (a+1)^2-(b^2-6b+9)$$

$$= (a+1)^2-(b-3)^2$$

$$= (a+b-2)(a-b+4) \checkmark$$

simple examples of algebra problems involving factorisations:

$$\sqrt{111113^2-888888}$$

$$x = 111111$$

$$\sqrt{(x+2)^2-8x}$$

$\sqrt{(x+2)^2 - 8x}$

$= \sqrt{x^2+4x+4-8x}$

$= \sqrt{x^2-4x+4}$ FACTORISE!!

$= \sqrt{(x-2)^2}$

$= x-2$

$= 111109$

$a^2 - b^2 = 20 \Rightarrow (a-b)(a+b) = 20$

$a - b = 4$

$a + b = ?$

$4(a+b) = 20$

$a+b = 5$ ✓

factorisations are also extremely useful in number theory